THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET 1

Chapter 14: **Probability** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	(d) $\frac{5}{36}$	Favourable outcomes: (2,6), (3,5), (4,4), (5,3), (6,2); number $= 5$; $P = \frac{5}{36}$
2	(d) $\frac{3}{8}$	Favourable outcomes: HHT, HTH, THH; number $= 3$; $P = \frac{3}{8}$
3	(d) $\frac{4}{11}$	Favourable letters: A, E, A, I; number $= 4$; $P = \frac{4}{11}$
4	(a) $\frac{1}{2}$	Favourable outcomes $=$ number of red balls; $P = \frac{6}{12} = \frac{1}{2}$
5	(b) $\frac{3}{10}$	Favourable numbers: 3, 6, 9, 12, 15, 18; number $= 6$; $P = \frac{6}{20} = \frac{3}{10}$
6	(d) $\frac{8}{9}$	Favourable outcomes $=$ number of good pens; $P = \frac{120}{135} = \frac{8}{9}$
7	(d) $\frac{1}{6}$	Favourable outcomes: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1); number $= 6$; $P = \frac{6}{36} = \frac{1}{6}$
8	(a) $\frac{1}{2}$	Favourable outcomes: HHH, HHT, HTH, THH; number $= 4$; $P = \frac{4}{8} = \frac{1}{2}$
9	(c) $\frac{2}{11}$	Favourable letters: B, B; number $= 2$; $P = \frac{2}{11}$
10	(a) $\frac{2}{3}$	Favourable outcomes $=$ number of red balls; $P = \frac{10}{15} = \frac{2}{3}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Total outcomes $= 2^3 = 8$: HHH, HHT, HTH, HTT, THH, THT, TTH, TTT; Favourable outcomes: HHT, HTH, HTT, THH, THT, TTH, TTT; number $= 7$ (1); $P = \frac{7}{8}$ (1).	1+1
12	Total number of outcomes $= 6 \times 6 = 36$; Favourable outcomes: (1,1), (1,2), (1,3), (2,1), (2,2), (3,1); number $= 6$ (1); $P = \frac{6}{36} = \frac{1}{6}$ (1).	1+1
13	Number of letters in the word $= 9$; Favourable letters: E, U, A, I, O; number $= 5$ (1); $P = \frac{5}{9}$ (1).	1+1
14	Total outcomes $= 2^3 = 8$: HHH, HHT, HTH, HTT, THH, THT, TTH, TTT; Favourable outcomes: HHH, HHT, HTH, HTT; number $= 4$ (1); $P = \frac{4}{8} = \frac{1}{2}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: (1,1), (1,4), (2,2), (3,3), (4,1), (4,4), (5,5), (6,6); number $= 8$ (1); $P = \frac{8}{36} = \frac{2}{9}$ (1).	1+1+1
16	Total possible outcomes $= 18$ (1); Favourable numbers: 3, 6, 9, 12, 15, 18; number $= 6$ (1); $P = \frac{6}{18} = \frac{1}{3}$ (1).	1+1+1
17	After adding, red balls $= 3 + 3 = 6$ and total balls $= 15$ (1); Favourable outcomes $=$ number of red balls (1); $P = \frac{6}{15} = \frac{2}{5}$ (1).	1+1+1
	OR: Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: (1,3), (2,2), (2,6), (3,1), (3,5), (4,4), (5,3), (6,2), (6,6); number $= 9$ (1); $P = \frac{9}{36} = \frac{1}{4}$ (1).	1+1+1
18	Total possible outcomes $= 28$ (1); Favourable numbers: 2, 3, 5, 7, 11, 13, 17, 19, 23; number $= 9$ (1); $P = \frac{9}{28}$ (1).	1+1+1
	OR: After adding, red balls $= 7 + 5 = 12$ and total balls $= 17$ (1); Favourable outcomes $=$ number of red balls (1); $P = \frac{12}{17}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total outcomes $= 36$ (2); Favourable outcomes: 5; 6; 4 (2); Probabilities: $\frac{5}{36}$, $\frac{1}{6}$, $\frac{1}{9}$ (1).	2+2+1
	OR: $\frac{4+x}{12+x} = \frac{1}{2}$ (2); Cross-multiplying: $2(4+x) = 1(12+x)$ (2); $x = 4$ (1).	2+2+1
20	Total outcomes $= 36$ (2); Favourable outcomes: 6; 6; 6 (2); Probabilities: $\frac{1}{6}$, $\frac{1}{6}$, $\frac{1}{6}$ (1).	2+2+1
	OR: $\frac{5+x}{12+x} = \frac{8}{15}$ (2); Cross-multiplying: $15(5+x) = 8(12+x)$ (2); $x = 3$ (1).	2+2+1

Mistakes to avoid:

- Writing a probability greater than 1 or negative; a probability always lies between 0 and 1.
- Using outcomes that are not equally likely (e.g. treating 'sum 7' and 'sum 12' as equally likely).
- Counting 1 as a prime number, or leaving out 2, the only even prime.
- Counting the overlap twice in 'A or B' questions.

- For two dice, using only 12 or 6 outcomes instead of 36, or counting (2,6) and (6,2) as the same outcome.
- Not changing the total number of outcomes when balls are removed or added, and not simplifying the fraction.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 14: **Probability** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	(b) $\frac{1}{5}$	Favourable numbers: 1, 4, 9, 16, 25; number $= 5$; $P = \frac{5}{25} = \frac{1}{5}$
2	(b) $\frac{8}{9}$	Favourable outcomes $=$ number of good pens; $P = \frac{80}{90} = \frac{8}{9}$
3	(a) $\frac{1}{6}$	Favourable outcomes: (1,1), (2,2), (3,3), (4,4), (5,5), (6,6); number $= 6$; $P = \frac{6}{36} = \frac{1}{6}$
4	(c) $\frac{1}{4}$	Favourable outcomes: HHH, TTT; number $= 2$; $P = \frac{2}{8} = \frac{1}{4}$
5	(a) $\frac{5}{11}$	Favourable letters: X, M, N, T, N; number $= 5$; $P = \frac{5}{11}$
6	(c) $\frac{10}{17}$	Favourable outcomes $=$ number of red balls; $P = \frac{10}{17}$
7	(a) $\frac{1}{3}$	Favourable numbers: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29; number $= 10$; $P = \frac{10}{30} = \frac{1}{3}$
8	(c) $\frac{21}{25}$	Favourable outcomes $=$ number of good pens; $P = \frac{63}{75} = \frac{21}{25}$
9	(c) $\frac{1}{9}$	Favourable outcomes: (2,6), (3,4), (4,3), (6,2); number $= 4$; $P = \frac{4}{36} = \frac{1}{9}$
10	(a) $\frac{1}{2}$	Favourable outcomes: HHH, HHT, HTH, THH; number $= 4$; $P = \frac{4}{8} = \frac{1}{2}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Total number of outcomes $= 6 \times 6 = 36$; Favourable outcomes: (1,3), (2,4), (3,1), (3,5), (4,2), (4,6), (5,3), (6,4); number $= 8$ (1); $P = \frac{8}{36} = \frac{2}{9}$ (1).	1+1
12	Number of letters in the word $= 7$; Favourable letters: S, S, S; number $= 3$ (1); $P = \frac{3}{7}$ (1).	1+1
13	Total outcomes $= 2^3 = 8$: HHH, HHT, HTH, HTT, THH, THT, TTH, TTT; Favourable outcomes: HHH, HTH; number $= 2$ (1); $P = \frac{2}{8} = \frac{1}{4}$ (1).	1+1
14	Total number of outcomes $= 6 \times 6 = 36$; Favourable outcomes: (1,1), (1,3), (1,5), (3,1), (3,3), (3,5), (5,1), (5,3), (5,5); number $= 9$ (1); $P = \frac{9}{36} = \frac{1}{4}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: 15 pairs; number $= 15$ (1); $P = \frac{15}{36} = \frac{5}{12}$ (1).	1+1+1
16	Total possible outcomes $= 40$ (1); Favourable numbers: 4, 8, 12, 16, 20, 24, 28, 32, 36, 40; number $= 10$ (1); $P = \frac{10}{40} = \frac{1}{4}$ (1).	1+1+1
17	After adding, red balls $= 4 + 4 = 8$ and total balls $= 16$ (1); Favourable outcomes $=$ number of red balls (1); $P = \frac{8}{16} = \frac{1}{2}$ (1).	1+1+1
	OR: Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: (6,6); number $= 1$ (1); $P = \frac{1}{36}$ (1).	1+1+1
18	Total possible outcomes $= 60$ (1); Favourable numbers: 6, 12, 18, 24, 30, 36, 42, 48, 54, 60; number $= 10$ (1); $P = \frac{10}{60} = \frac{1}{6}$ (1).	1+1+1
	OR: After adding, red balls $= 8 + 2 = 10$ and total balls $= 16$ (1); Favourable outcomes $=$ number of red balls (1); $P = \frac{10}{16} = \frac{5}{8}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total outcomes $= 36$ (2); Favourable outcomes: 8; 9; 15 (2); Probabilities: $\frac{2}{9}$, $\frac{1}{4}$, $\frac{5}{12}$ (1).	2+2+1
	OR: $\frac{3+x}{12+x} = \frac{4}{7}$ (2); Cross-multiplying: $7(3+x) = 4(12+x)$ (2); $x = 9$ (1).	2+2+1
20	Total outcomes $= 36$ (2); Favourable outcomes: 8; 9; 15 (2); Probabilities: $\frac{2}{9}$, $\frac{1}{4}$, $\frac{5}{12}$ (1).	2+2+1
	OR: $\frac{6+x}{12+x} = \frac{2}{3}$ (2); Cross-multiplying: $3(6+x) = 2(12+x)$ (2); $x = 6$ (1).	2+2+1

Mistakes to avoid:

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- Counting the overlap twice in 'A or B' questions.
- For two dice, using only 12 or 6 outcomes instead of 36, or counting (2,6) and (6,2) as the same outcome.
- Not changing the total number of outcomes when balls are removed or added, and not simplifying the fraction.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 14: **Probability** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	(b) $\frac{4}{13}$	Favourable letters: S, S, S, S; number $= 4$; $P = \frac{4}{13}$
2	(a) $\frac{5}{7}$	Favourable outcomes = number of red balls; $P = \frac{10}{14} = \frac{5}{7}$
3	(c) $\frac{1}{6}$	Favourable numbers: 6, 12, 18, 24; number $= 4$; $P = \frac{4}{24} = \frac{1}{6}$
4	(c) $\frac{22}{25}$	Favourable outcomes = number of good pens; $P = \frac{176}{200} = \frac{22}{25}$
5	(d) $\frac{1}{6}$	Favourable outcomes: (4,6), (5,5), (5,6), (6,4), (6,5), (6,6); number $= 6$; $P = \frac{6}{36} = \frac{1}{6}$
6	(b) $\frac{3}{8}$	Favourable outcomes: HTT, THT, TTH; number $= 3$; $P = \frac{3}{8}$
7	(a) $\frac{2}{9}$	Favourable letters: E, E; number $= 2$; $P = \frac{2}{9}$
8	(b) $\frac{5}{13}$	Favourable outcomes = number of red balls; $P = \frac{5}{13}$
9	(d) $\frac{1}{5}$	Favourable numbers: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50; number $= 10$; $P = \frac{10}{50} = \frac{1}{5}$
10	(d) $\frac{19}{20}$	Favourable outcomes = number of good pens; $P = \frac{152}{160} = \frac{19}{20}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Number of letters in the word $= 9$; Favourable letters: M, M; number $= 2$ (1); $P = \frac{2}{9}$ (1).	1+1
12	Total outcomes $= 2^3 = 8$: HHH, HHT, HTH, HTT, THH, THT, TTH, TTT; Favourable outcomes: HHT, HTH, HTT, THH, THT, TTH, TTT; number $= 7$ (1); $P = \frac{7}{8}$ (1).	1+1
13	Total number of outcomes $= 6 \times 6 = 36$; Favourable outcomes: 15 pairs; number $= 15$ (1); $P = \frac{15}{36} = \frac{5}{12}$ (1).	1+1
14	Number of letters in the word $= 12$; Favourable letters: T, R, G, N, M, T, R, Y; number $= 8$ (1); $P = \frac{8}{12} = \frac{2}{3}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: (1,4), (2,3), (3,2), (4,1); number $= 4$ (1); $P = \frac{4}{36} = \frac{1}{9}$ (1).	1+1+1
16	Total possible outcomes $= 32$ (1); Favourable numbers: 1, 4, 9, 16, 25; number $= 5$ (1); $P = \frac{5}{32}$ (1).	1+1+1
17	After adding, red balls $= 5 + 5 = 10$ and total balls $= 20$ (1); Favourable outcomes = number of red balls (1); $P = \frac{10}{20} = \frac{1}{2}$ (1).	1+1+1
	OR: Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: (3,6), (4,5), (5,4), (6,3); number $= 4$ (1); $P = \frac{4}{36} = \frac{1}{9}$ (1).	1+1+1
18	Total possible outcomes $= 35$ (1); Favourable numbers: 5, 10, 15, 20, 25, 30, 35; number $= 7$ (1); $P = \frac{7}{35} = \frac{1}{5}$ (1).	1+1+1
	OR: After adding, red balls $= 9 + 6 = 15$ and total balls $= 18$ (1); Favourable outcomes = number of red balls (1); $P = \frac{15}{18} = \frac{5}{6}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total outcomes $= 36$ (2); Favourable outcomes: 1; 4; 4 (2); Probabilities: $\frac{1}{36}, \frac{1}{9}, \frac{1}{9}$ (1).	2+2+1
	OR: $\frac{2+x}{12+x} = \frac{3}{8}$ (2); Cross-multiplying: $8(2+x) = 3(12+x)$ (2); $x = 4$ (1).	2+2+1
20	Total outcomes $= 36$ (2); Favourable outcomes: 9; 11; 4 (2); Probabilities: $\frac{1}{4}, \frac{11}{36}, \frac{1}{9}$ (1).	2+2+1
	OR: $\frac{7+x}{12+x} = \frac{16}{21}$ (2); Cross-multiplying: $21(7+x) = 16(12+x)$ (2); $x = 9$ (1).	2+2+1

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